SENG 300 – Leaderboard Subsystem
Rules & Validation + Quality

Project kickoff presentation

SECTION 8

Contents

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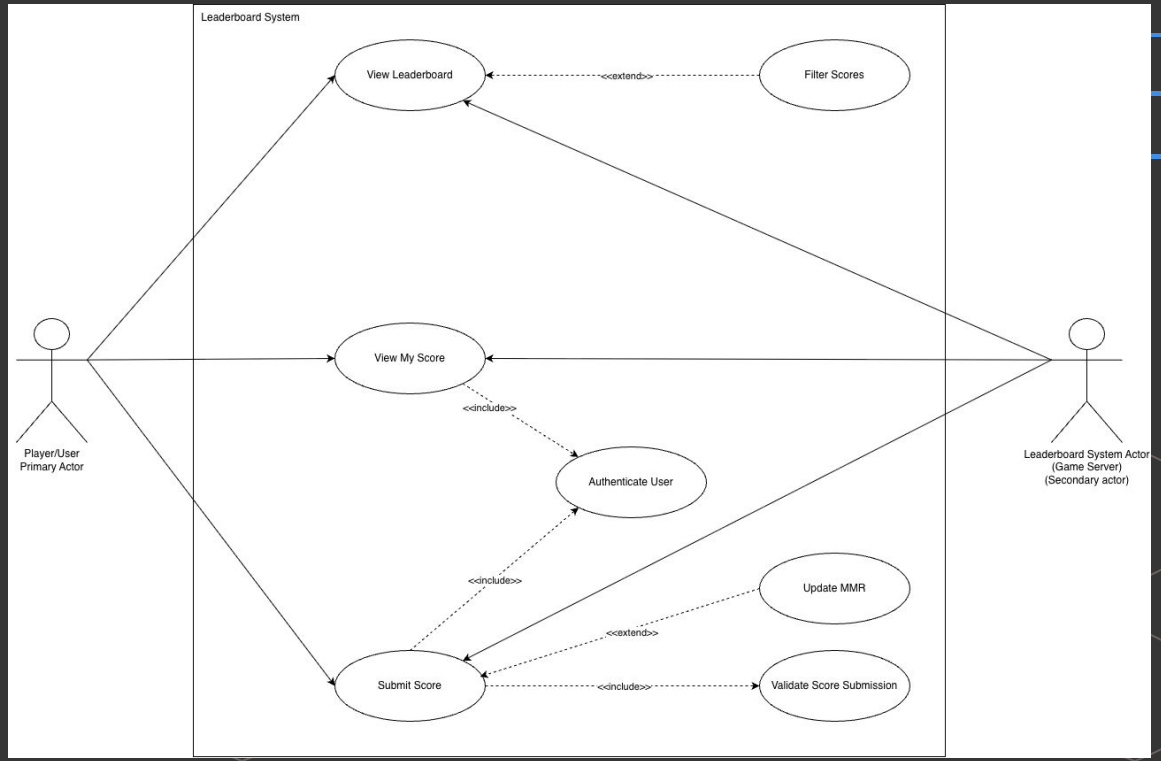
USE CASE DIAGRAM

User and Server Use Cases:

- View Leaderboard
- View My Score
- Submit Score
- Authenticate User
- Validate Score Submission
- Filter Scores
- Update MMR



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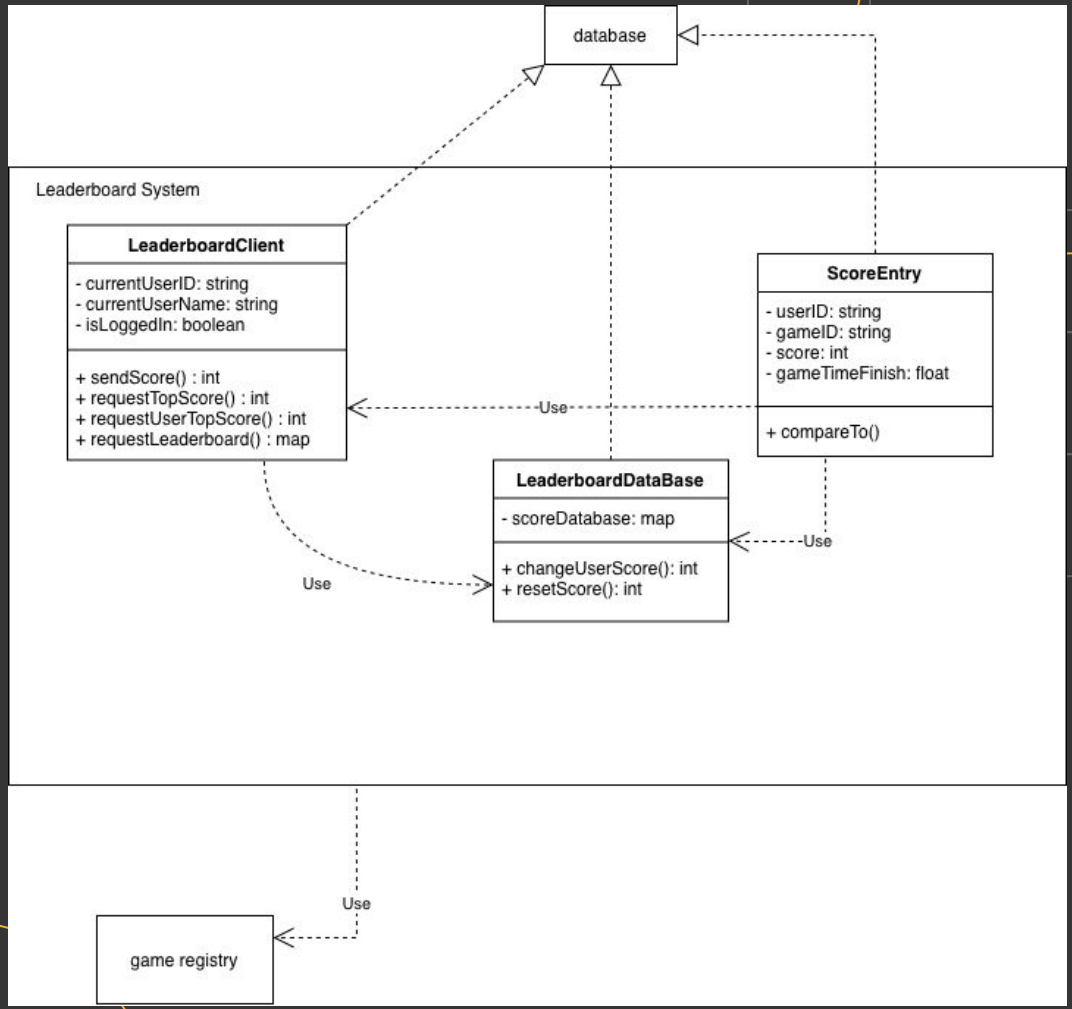
CLASS DIAGRAM

Classes and External Dependencies

- LeaderboardClient
- LeaderboardDatabase
- ScoreEntry
- Database *(external dependency)*
- Game Registry *(external dependency)*



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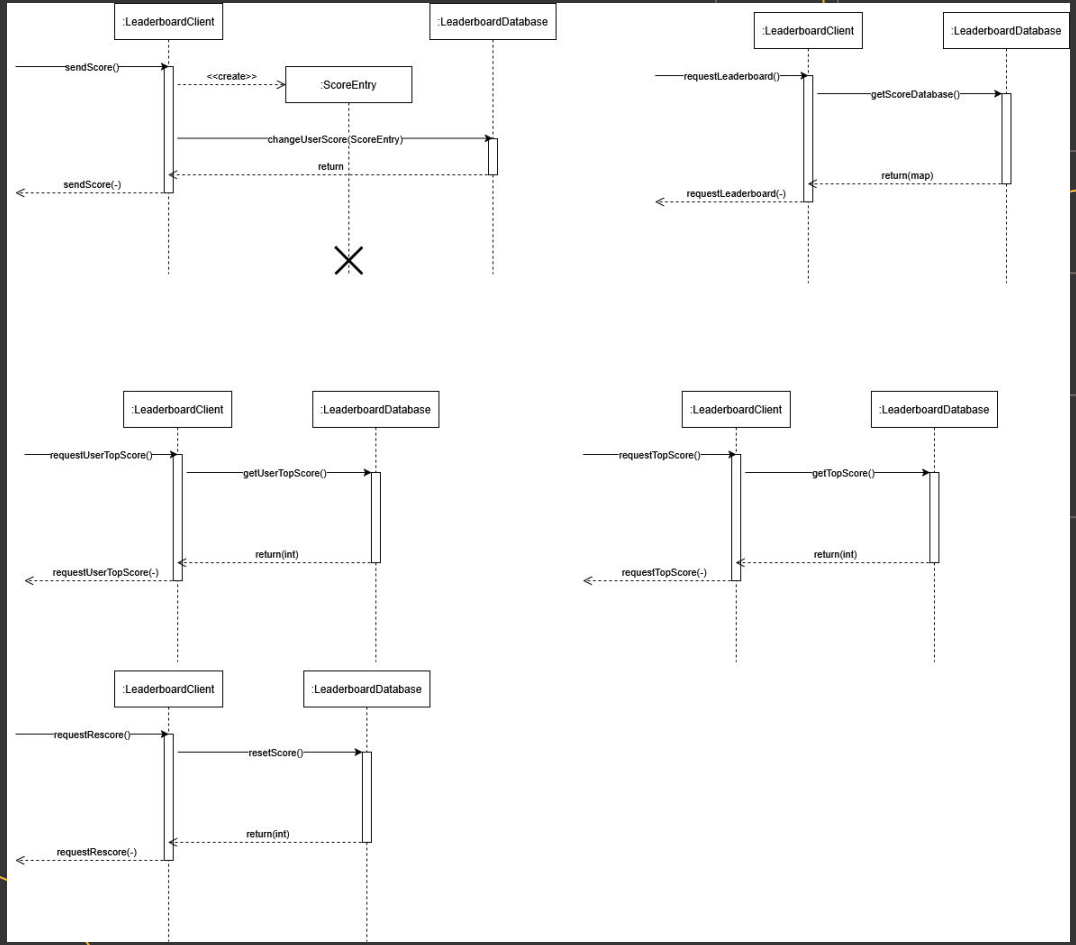
SEQUENCE DIAGRAM



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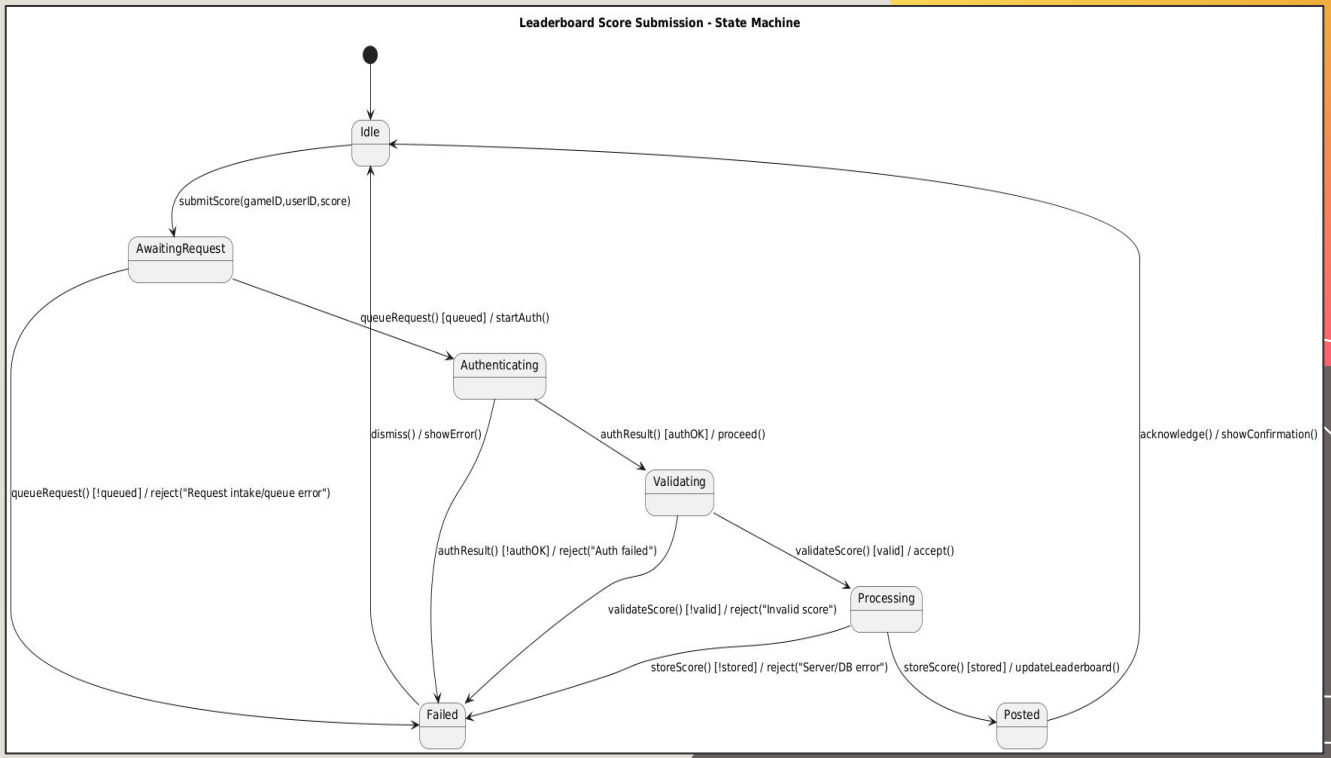
STATE MACHINE DIAGRAM SCORE SUBMISSION LIFECYCLE

This diagram models the lifecycle of a score submission from initiation to completion or failure.

- Idle
- Authenticating
- Validating
- Processing
- Posted
- Failed

Separating authentication and validation ensures identity verification and rule compliance remain distinct responsibilities

Any failure transitions to a Failed state, ensuring no invalid data reaches persistence.



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DESIGN RATIONALE

Server-side validation → Prevents client-side tampering and ensures all score submissions are verified before reaching persistence.

Layered architecture → Separates client, server, and database responsibilities to improve modularity, scalability, and maintainability.

Separation of concerns →Authentication, validation, processing, and storage are handled in distinct components to reduce coupling and improve clarity.

Controlled failure states →All invalid or error conditions transition to a defined “Failed” state, ensuring system integrity and preventing corrupted data.

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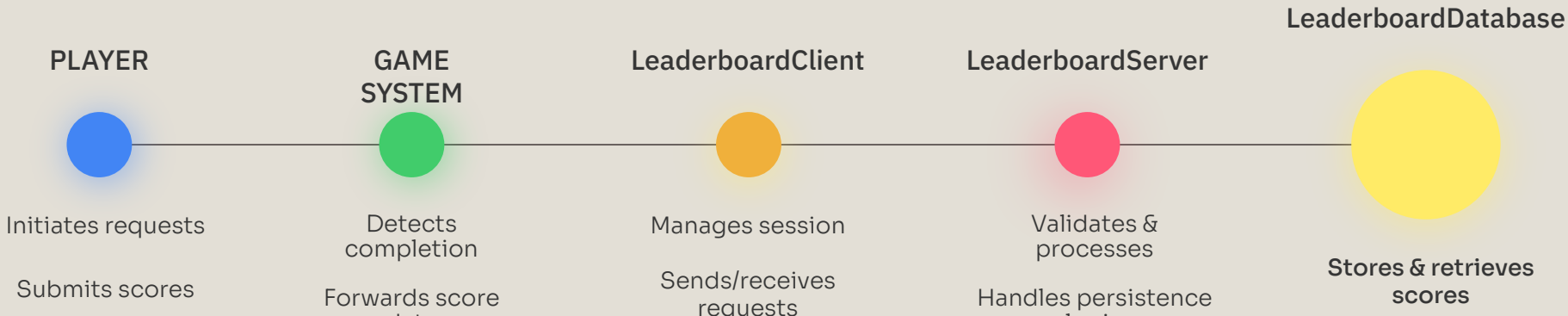
INTEGRATION



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DESIGN CRITIQUE

- Starter documentation lacked explicit failure modeling
- Authentication and validation responsibilities were not clearly separated
- Client–server boundaries were loosely defined
- Submission lifecycle was not fully modeled
- Persistence assumptions were ambiguous

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CHALLENGES IDENTIFIED

- Interpreting unclear starter design assumptions
- Establishing strict responsibility boundaries between system layers
- Modeling safe and deterministic failure transitions
- Aligning integration contracts with other sub-teams
- Managing design depth within iteration constraints

Moving Forward

Step 1

Finalize subsystem interfaces and integration points

Step 2

Implement server-side validation and processing logic

Step 3

Complete database storage and retrieval mechanisms

Step 4

Perform end-to-end integration and testing

Step 5

Optimize for performance and scalability

Goal

Deliver a secure, modular, and production-ready leaderboard subsystem

Thank you

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